

DSA0601 • DATA HANDLING AND VISUALIZATION LAB EXPERIMENT RECORD

Set 13: Geographic Data Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Aim

To analyze geographic data using a scatter plot and tabular representation, and to develop a simple Tableau dashboard combining the visualizations for interactive exploration.

Dataset: Geographic Data

City	Population	Avg. Temperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.
2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.
3. Build a table to display the geographic data for each city. Label the table elements.
4. Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

PROGRAM 1: Geographic Data and Scatter Plot

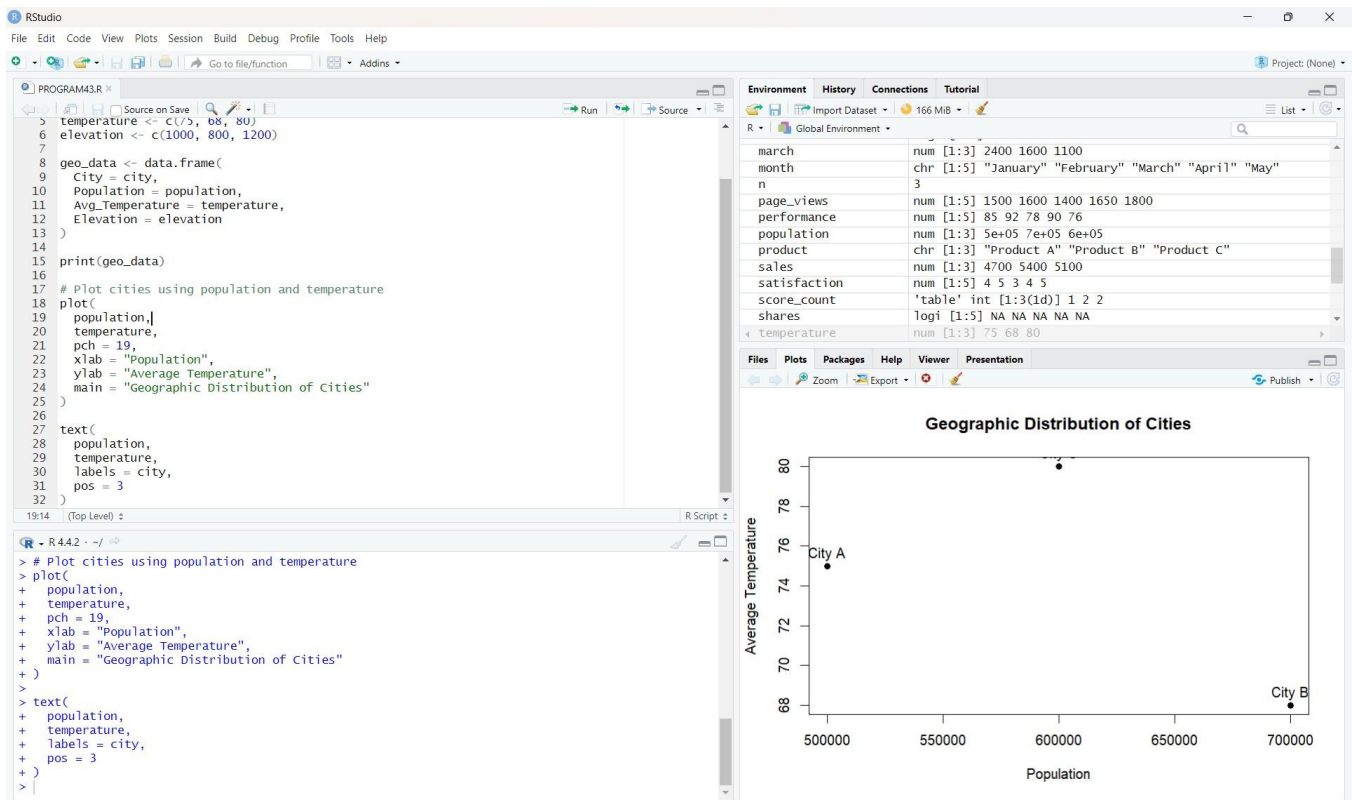
Create the geographic data and visualize population against average temperature. Label the plotted cities.

```
# Program 1: Geographic Data and Scatter Plot
city <- c("City A", "City B", "City C")
population <- c(500000, 700000, 600000)
temperature <- c(75, 68, 80)
elevation <- c(1000, 800, 1200)

geo_data <- data.frame(
  City = city,
  Population = population,
  Avg_Temperature = temperature,
  Elevation = elevation
)
print(geo_data)

plot(population, temperature, pch = 19,
     xlab = "Population", ylab = "Average Temperature",
     main = "Geographic Distribution of Cities")
text(population, temperature, labels = city, pos = 3)
```

Output 1: Geographic Distribution of Cities



Observation: The plot shows City A, City B, and City C according to their population and average temperature values.

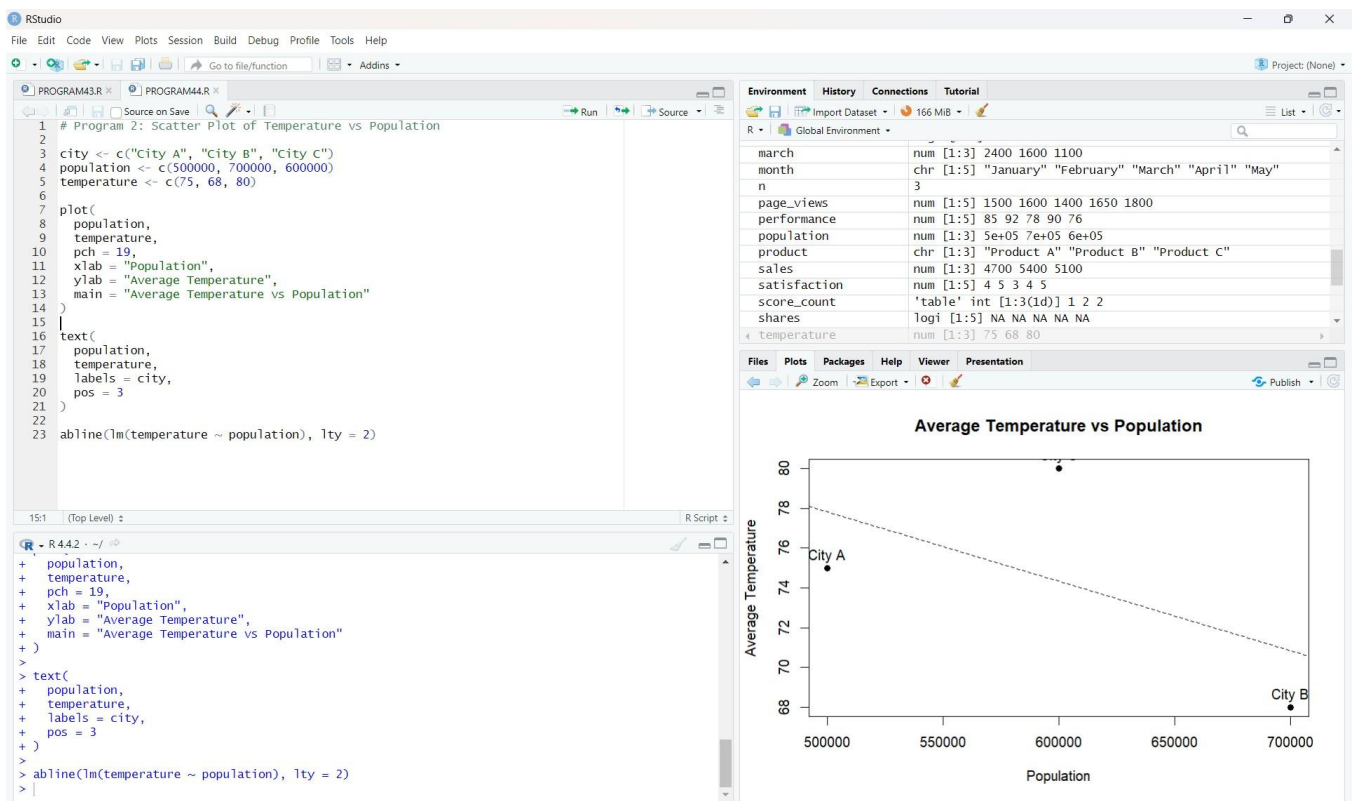
PROGRAM 2: Scatter Plot of Temperature vs Population

Generate a scatter plot to explore the relationship between average temperature and population size.

```
# Program 2: Scatter Plot of Temperature vs Population
city <- c("City A", "City B", "City C")
population <- c(500000, 700000, 600000)
temperature <- c(75, 68, 80)

plot(population, temperature, pch = 19,
      xlab = "Population", ylab = "Average Temperature",
      main = "Average Temperature vs Population")
text(population, temperature, labels = city, pos = 3)
abline(lm(temperature ~ population), lty = 2)
```

Output 2: Average Temperature vs Population



Output 2: Scatter Plot of Temperature vs Population

Observation: City C has the highest average temperature (80), while City B has the largest population (700000) and the lowest temperature (68).

PROGRAM 3: Geographic Data Table

Build a table to show the geographic data for each city. Label the table elements.

```
# Program 3: Geographic Data Table
geo_data <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Avg_Temperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200)
)
print(geo_data)
```

Output 3: Geographic Data Table

The screenshot shows the RStudio interface with the following components:

- Source Editor:** Contains the R script for Program 3: Geographic Data Table.
- Console:** Displays the execution output, showing the creation of the `geo_data` data frame and its printed representation.
- Environment Pane:** Lists the objects in the global environment, including `geo_data`, `interactions`, `sales_data`, and `ui`.
- Files Pane:** Shows the project structure, including the `PROGRAM3.R` file.

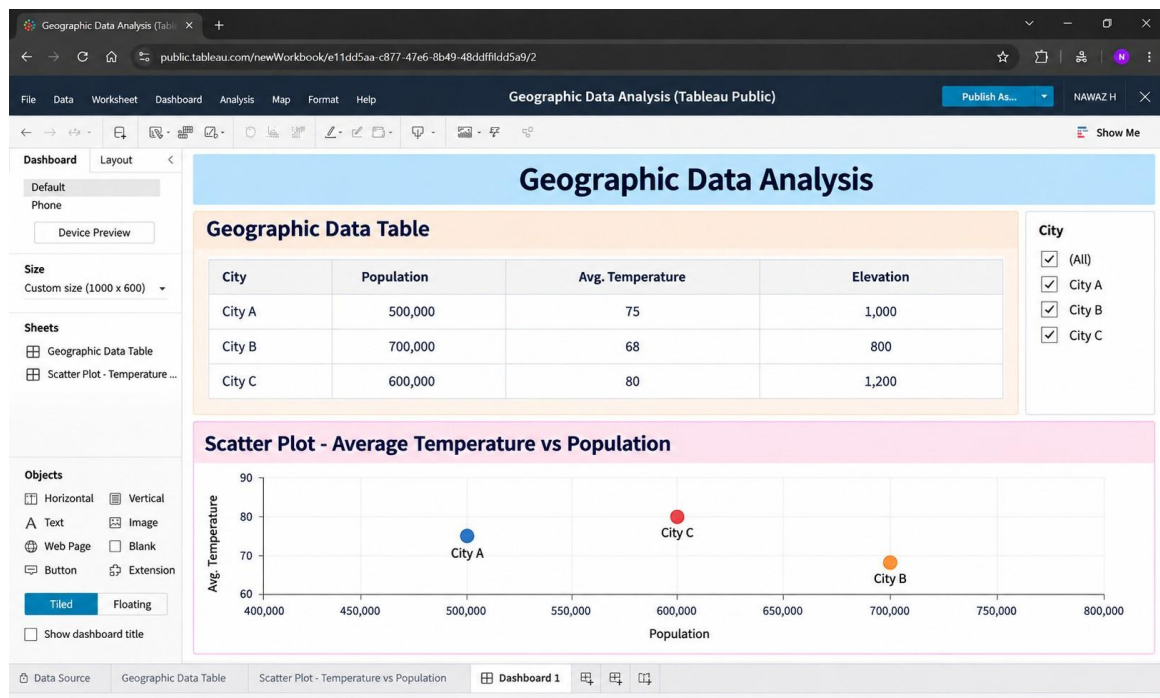
The console output is as follows:

```
> abline(lm(temperature ~ population), lty = 2)
> # Program 3: Geographic Data Table
>
> geo_data <- data.frame(
+   City = c("City A", "City B", "City C"),
+   Population = c(500000, 700000, 600000),
+   Avg_Temperature = c(75, 68, 80),
+   Elevation = c(1000, 800, 1200)
+ )
>
> print(geo_data)
  City Population Avg_Temperature Elevation
1 City A      5e+05          75      1000
2 City B      7e+05          68       800
3 City C      6e+05          80      1200
>
```

Observation: The table displays the three cities with their population, average temperature, and elevation values.

PROGRAM 4: Tableau Dashboard

Develop a Tableau dashboard combining the geographic data table and scatter plot for interactive exploration.



Output 4: Tableau Geographic Data Analysis Dashboard

Observation: The Tableau dashboard combines the geographic data table and the average temperature versus population scatter plot. The City filter can be used to explore the displayed data interactively.

Procedure

1. Create the city, population, temperature, and elevation vectors in R.
2. Create a data frame containing the geographic information.
3. Use `plot()` and `text()` to generate and label the scatter plot.
4. Use `abline()` to display the linear trend in the temperature versus population plot.
5. Create and print the geographic data table using `data.frame()`.
6. Enter the geographic data into Tableau and create the required worksheets.
7. Combine the table and scatter plot into a dashboard and add a City filter for interactive exploration.

Result

The geographic data was successfully visualized using a scatter plot and table in R. A simple Tableau dashboard was also created by combining the geographic data table and scatter plot for interactive exploration.

Key R Functions Used

Function	Purpose
<code>data.frame()</code>	Creates the geographic data table.
<code>plot()</code>	Creates the scatter plot.
<code>text()</code>	Labels the plotted cities.
<code>abline()</code>	Adds the linear trend line.
<code>print()</code>	Displays the geographic data.